

# Rishi More

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## Summary

ML engineer building **risk-controlled LLM inference**, **retrieval systems**, and **production-grade ML pipelines**. Co-first author of an **ICML 2026** paper on distribution-free risk control for reducing test-time compute under calibrated error guarantees. Experienced translating research prototypes into reliable, low-latency ML systems.

## Experience

### JHU Data Science and AI Institute (DSAI)

May 2025 – Present

Graduate Research Assistant (Advisors: Prof. Daniel Khashabi, Prof. Eric Nalisnick)

Baltimore, MD

- **Co-first author** of *Conformal Thinking (ICML 2026)*: distribution-free risk control that sets dual confidence thresholds for adaptive reasoning, cutting LLM inference cost while bounding error at a user-specified risk level.
- Ran vLLM experiments across **4** reasoning models and **4** benchmarks (**2.7K+** problems), maintained target error rates under dataset/length shift using finite-sample (UCB) calibration.
- Designed a parametric lower threshold and ensembled stopping rule, cutting reasoning tokens **12%** at matched accuracy; built reusable calibration/evaluation tooling.

### JHU Center for Language and Speech Processing (CLSP)

January 2025 – December 2025

Graduate Researcher (Advisor: Prof. Jason Eisner)

Baltimore, MD

- Built a temporal-aware RAG pipeline over **10K+** non-stationary documents using time-decay weighting and metadata-aware indexing.
- Raised retrieval precision **23%** over dense baselines by surfacing temporally relevant evidence in high-volatility knowledge domains.
- Cut false retrievals **31%** with a reference-free consistency-scoring framework for hallucination detection.

### Mastek

November 2022 – July 2023

Machine Learning Engineer Intern

Mumbai, India

- Built and deployed an on-prem **edge** voice authentication system at **<100ms p99** latency and **96.6%** accuracy, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment under edge-compute limits.
- Engineered a TensorFlow pipeline over **1M+** voice samples with spectral-gating denoising for robust feature extraction.
- Improved authentication reliability **20%** in high-noise production environments via preprocessing and feature-extraction upgrades.

## Projects

### Post-Training Evaluation & Reward-Hacking Detector | Python, PyTorch, Hugging Face, DPO, LLM Judges, Calibration | GitHub

- Built an evaluation harness for **6** post-training failure modes—reward hacking, sycophancy, verbosity bias, false confidence, safety drift, and distribution shift—across **4** Qwen3-8B variants: base, SFT, DPO, and instruct.
- Evaluated **142** adversarial prompts with debiased pairwise LLM judging, position-swapped comparisons, and log-prob calibration, flagging **10%** of DPO preference wins that degraded factuality.
- Attributed regressions to optimization stages using ECE, Brier score, AUROC, Wilson/bootstrap confidence intervals, and **28** offline tests; found SFT introduced safety drift and DPO increased sycophancy from **34%** to **45%**.

### Asymmetric Relation Dynamics in Language Agents | Python, LLM Agents, LoRA, AsyncIO, BM25 | GitHub

- Built an async multi-agent training system (Tinker API): a frozen **Qwen3-235B** caregiver teaching LoRA-adapted **Qwen3-8B** agents via a four-module memory and salience-gated BM25 retrieval.
- Reached **100%** training success with **~25%** fewer turns (6.1 vs 8.4) over Solo/Peer baselines across **4** conditions  $\times$  **3** seeds.
- Isolated scaffolding dependency as the transfer bottleneck via ablations; fully reproducible via a single CLI command.

### Label-Preserving Domain Adaptation via KL-Regularized Optimal Transport | Python, POT/Sinkhorn, K-Means, t-SNE | GitHub

- Extended Sinkhorn OT with a KL-divergence information loss and intra-cluster consistency penalty, preserving label structure under unsupervised domain adaptation without target labels.
- Cut transport cost from  $O(n_s^2)$  to  $O(n_c^2)$  via centroid-level cluster transport.
- Improved 1-NN target accuracy over Sinkhorn by **+12.6pp** on Two-Moons (99.8%) and **+5.3pp** on MNIST→USPS (82.7%).

## Education

### Johns Hopkins University

May 2026

M.S.E. in Computer Science

GPA: 4.0/4.0

- **Graduate coursework:** Theory of Replicable ML, Human-in-the-Loop ML, Non-Stationary Environments.

### University of Mumbai

July 2024

B.Tech. in Computer Science, Honors in AI/ML

CGPA: 9.85/10

## Technical Skills

**Languages:** Python, C++, Java, C

**ML/LLM Systems:** PyTorch, TensorFlow, Hugging Face, LoRA, DPO, RAG, calibration, conformal/risk control, Scikit-learn

**Inference & MLOps:** vLLM, Docker, Kubernetes, AWS, Apache Airflow, Apache Spark

**Data:** SQL, PostgreSQL, MongoDB, Pandas, NumPy